

# Approximate Bayesian inference of gene-family coupling

Fitting a likelihood-free Potts model to phylogenetic profiles

ZOMBI2 • coupling ABC analysis

simulated-data validation

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## Abstract

We ask whether the *strength* of gene-family coupling can be recovered from phylogenetic profiles by Approximate Bayesian Computation (ABC) — a step toward simulating realistic synthetic datasets from empirical ones (e.g. eggNOG) under a model whose likelihood is intractable. On simulated data, ABC recovers the coupling strength  $J$  accurately (RMSE 0.056). We identify a key entanglement: coupling *sustains* gene families, so it leaves a strong *marginal* (prevalence) footprint — which means the frequency spectrum alone already recovers  $J$  in ordinary regimes. We then stress-test the method by holding prevalence fixed, where the marginal is *blind*: there, gene-family *co-occurrence* structure becomes essential and roughly halves the error ( $0.35 \rightarrow 0.16\text{--}0.21$ ). This co-occurrence signal is the phylogenetic-profiling (Pellegrini) signal, and it stands clearly above the phylogenetic confound on a realistic tree. Finally, on a real eggNOG-annotated dataset we find that orthologous groups sharing a COG functional category co-occur significantly more than groups in different categories ( $p \approx 0.002$  against a phylogeny-controlled null), with mobile/accessory functions the most cohesive — support for using COG categories to scaffold the coupling structure.

## 1 The question

ZOMBI2 simulates a species tree and then evolves gene families along it. In the *coupled* (Potts) model, the presence/absence of a fixed panel of  $N$  families is an Ising vector  $\sigma$ , and a family's loss rate depends on its partners:

$$\text{loss}_i = \text{base\_loss} \cdot e^{-\beta f_i}, \quad f_i = h_i + \sum_j J_{ij} \sigma_j,$$

so a positive coupling  $J_{ij} > 0$  makes two families tend to *co-occur* (present partners lower each other's loss). Given an empirical profile — a families  $\times$  species copy-number table — we would like to *infer* this model, so we can generate synthetic datasets that look like the real one. Because the model includes transfer and coupling, its likelihood is not computable, so we use ABC (likelihood-free inference).

Here we validate the core question on simulated data, where the truth is known: can ABC recover the coupling *strength*  $J$  of a *known* pathway structure? (Inferring the structure itself — which families couple — is a separate, harder problem, left for later.)

## 2 Method

**The simulator.** We place  $N = 60$  families in 15 pathways of 4, couple every pair *within* a pathway with strength  $J$  (nothing between pathways), and simulate presence/absence along a 60-tip birth-death tree with `simulate_coupled`.

**ABC.** We draw  $J \sim \text{Uniform}(0, 1.5)$ , simulate a profile, reduce it to *summary statistics*, measure the (variance-scaled Euclidean) distance to the empirical summary, and keep the 5% of the 500 draws whose summary lands closest. The accepted  $J$  values are the posterior.

**Summary statistics.** Two kinds, both blind to family identity/order:

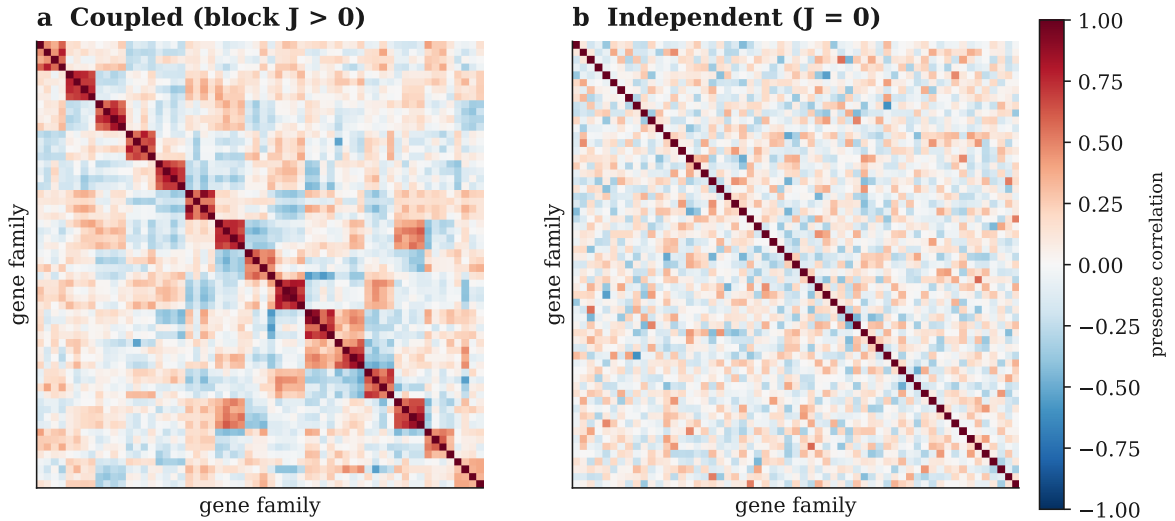
- *Marginal* — the gene *frequency spectrum* (how many families are present in exactly  $k$  species), plus genome sizes and a copy-number spectrum. This sees each family’s prevalence, but not whether families co-occur.
- *Co-occurrence* — structure of the family $\times$ family presence-correlation graph. We compare three: the correlation *histogram*, its leading *eigenvalues*, and *clustering* (edges, triangles, transitivity of the thresholded graph). Clustering targets *modules* — cliques of mutually co-occurring families — which is the biological signal.

### 3 What we see

#### 3.1 Co-occurrence carries the signal, above the confound

Coupling imprints *module structure* on the profiles: families in a pathway co-occur, so the presence-correlation matrix shows blocks (Fig. 1a) that are absent under independence (Fig. 1b). At matched prevalence, a coupled dataset has 128 triangles in its thresholded co-occurrence graph versus 1 for an independent one. Crucially, the *phylogenetic confound* — independent families co-occurring merely because they share ancestry — is mild on this tree (mean pairwise correlation  $\approx 0$  across prevalences), so the coupling signal is not an artifact of the tree.

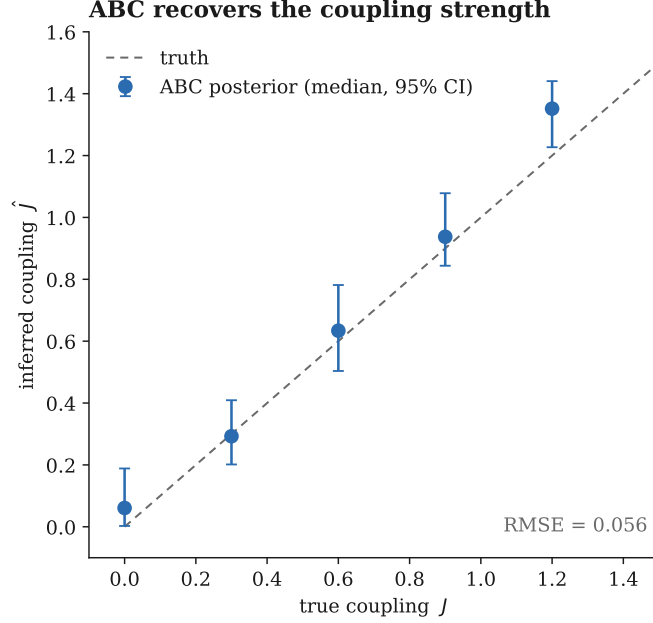
**Gene-family co-occurrence (both at prevalence  $\approx 0.5$ )**



**Figure 1.** Gene-family presence-correlation matrices at matched prevalence ( $\approx 0.5$ ). Coupling produces block modules (a); independence does not (b).

#### 3.2 ABC recovers the coupling strength

In an ordinary regime, the ABC posterior median tracks the true coupling across the whole range, with the truth inside every 95% credible interval and  $\text{RMSE} = 0.056$  (Fig. 2). The direction works end to end: coupling strength is identifiable from profiles.



**Figure 2.** Inferred vs. true coupling  $J$  (posterior median, 95% CI). Dashed line is the truth.

### 3.3 The entanglement — and when co-occurrence is essential

Why does Fig. 2 work *even* for the marginal summary? Because coupling and prevalence are **entangled**: coupling keeps families alive (co-retention), so a stronger  $J$  raises prevalence and leaves a strong footprint in the frequency spectrum. The marginal therefore recovers  $J$  *indirectly*, through abundance.

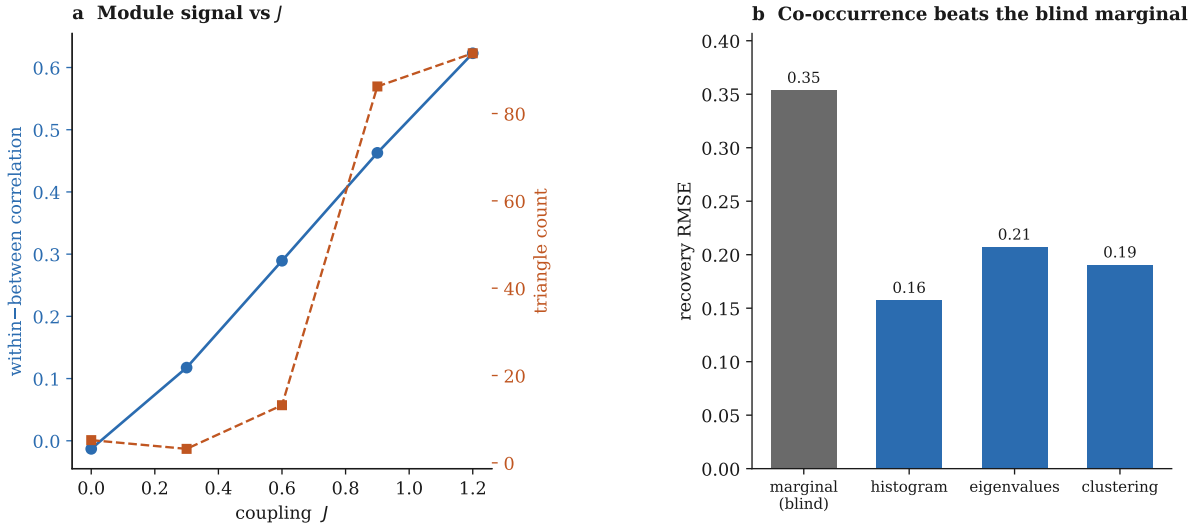
To test whether co-occurrence is *necessary*, we remove that shortcut: we calibrate the field  $h(J)$  to hold prevalence at  $\approx 0.5$  for every  $J$ , so the marginal goes blind. The module signal is undisturbed — the within-minus-between pathway correlation still climbs cleanly from 0 to 0.62 as  $J$  goes  $0 \rightarrow 1.2$  (Fig. 3a) — but the marginal summary now recovers nothing (RMSE 0.354, its posterior is essentially the prior). The co-occurrence summaries, in contrast, still recover  $J$  and roughly *halve* the error (Fig. 3b): histogram 0.157, clustering 0.190, eigenvalues 0.207. Figure 4 shows the payoff directly: clustering tracks  $J$  where the marginal is flat.

## 4 A first look at real data: eggNOG and the COG-coupling idea

We close by asking whether any of this is visible in a real dataset, and by testing a specific idea: *could the coupling structure be scaffolded by function* — families in the same COG functional category coupled to each other? That would turn the hardest open problem (inferring *which* families couple) into a principled prior.

**Data and parser.** We parse an eggNOG-mapper annotation of a set of bacterial genomes, collapsing genes to orthologous groups (the broadest, root-level OG) to build a profile of 4346 groups, and recording each group’s COG category. Its pangenome spectrum has the familiar U-shape — many genome-specific groups, a smaller universal core (Fig. 5a). One caveat sets the ceiling on power here: the mini annotation covers only 8 genomes, so per-pair co-occurrence is noisy; we therefore test only *aggregate* signals, averaged over thousands of group pairs.

**Same-category groups co-occur more.** Orthologous groups sharing a COG category co-occur more than groups in different categories. To guard against the phylogenetic confound, we compare against a *label-permutation null*: we keep every group’s real presence pattern (so any



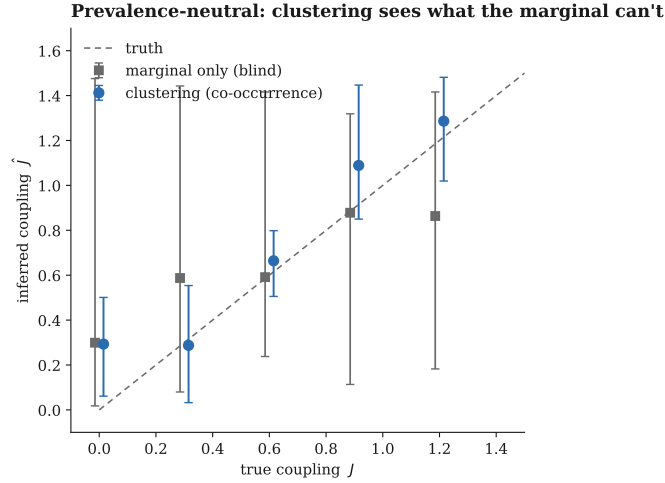
**Figure 3.** Prevalence held fixed at  $\approx 0.5$ . (a) The module signal grows with  $J$  even though prevalence does not. (b) Recovery error by summary (averaged over 8 target datasets per  $J$ ): every co-occurrence summary roughly halves the blind-marginal error.

tree-driven co-occurrence is present in both observed and null) and only shuffle the category labels. The observed within-minus-between gap ( $+0.009$ ) sits far outside the null ( $p \approx 0.002$ ,  $z \approx 3.6$ ; Fig. 5b) — evidence for *functional* structure beyond the tree.

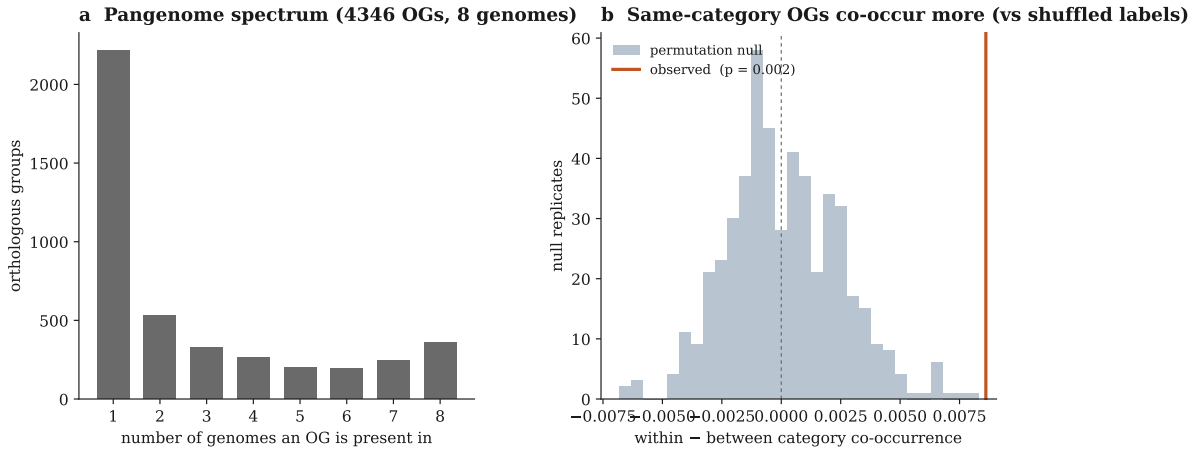
**Which functions are modular.** The signal is concentrated in the *accessory* / *mobile* functions (Fig. 6): cell motility is by far the most cohesive category (flagellar genes are a textbook co-inherited module), followed by carbohydrate metabolism, secretion, secondary metabolites, and defense — precisely the functions known to be gained and lost as units. Core housekeeping (translation, replication, energy) shows near-zero or negative cohesion, as expected: those genes are near-universal, so they barely co-vary. Functional categories thus capture real co-occurrence modules, and the biology of *which* ones is sensible — a strong hint that COG categories are a usable scaffold for the coupling matrix.

## 5 Takeaways and next steps

- **ABC recovers gene-family coupling from profiles** (RMSE 0.056) — the likelihood-free route works on simulated data.
- **Fit the marginal and the co-occurrence together.** Coupling shows in prevalence *and* in module structure; the marginal handles the common (prevalence-shifting) case, co-occurrence rescues the prevalence-neutral one. The two together are never worse than the marginal alone. We ship *clustering* (module structure) as the co-occurrence summary for its interpretability; the correlation histogram is a comparable alternative (the three co-occurrence summaries are close, with the histogram marginally ahead here).
- **Honest limits.** Faint coupling ( $J \lesssim 0.3$ ) at fixed prevalence is intrinsically hard — weak modules against finite-species estimation noise — so expect wide posteriors at low  $J$ . And the phylogenetic confound is *tree-dependent*: mild here, it must be re-checked on a real (deeper, clumpier) eggNOG tree.
- **Real data shows functional co-occurrence.** An eggNOG dataset already carries a significant same-COG-category co-occurrence signal, concentrated in the mobile/accessory functions. The *effect is small* at this sample size (only 8 genomes annotated) — enough to



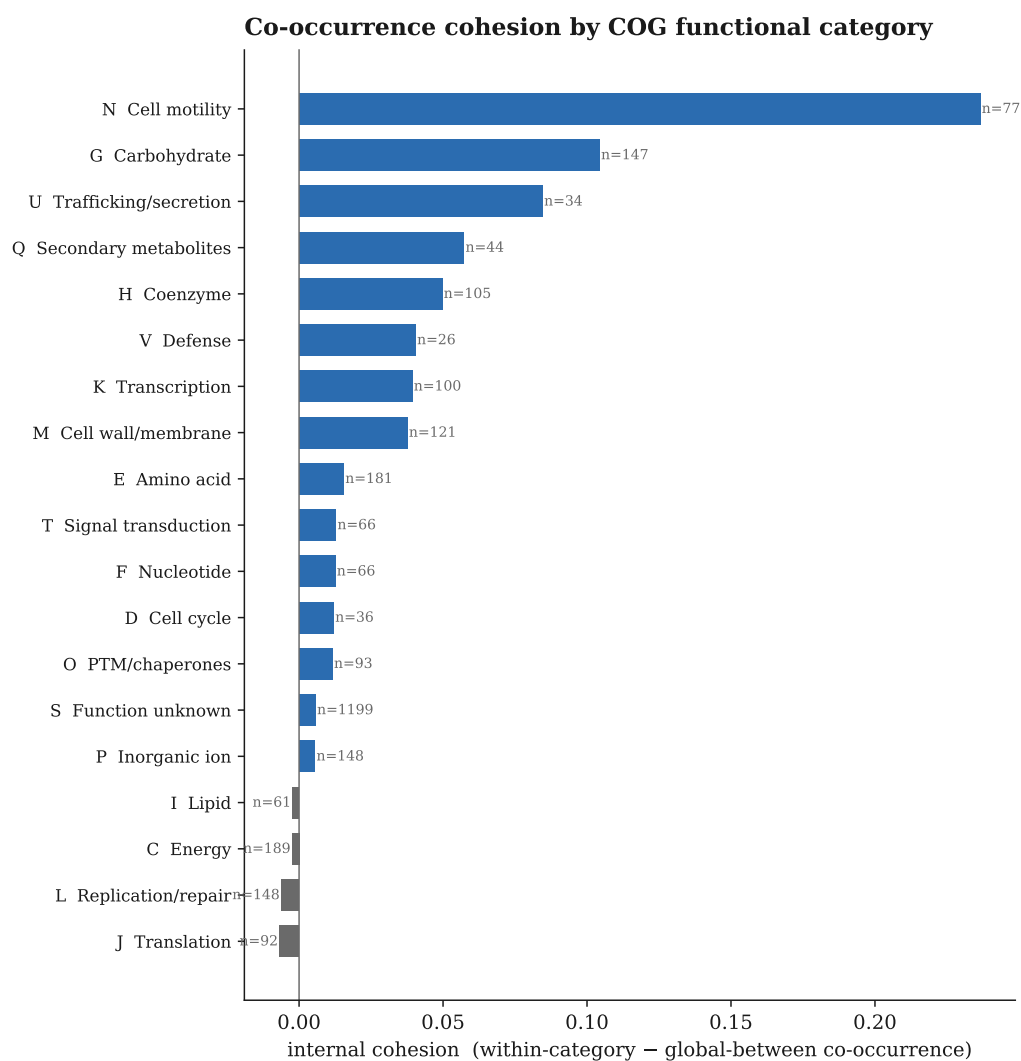
**Figure 4.** Prevalence-neutral recovery. The marginal (grey) is flat — blind to  $J$ ; the co-occurrence (clustering) summary (blue) tracks the truth.



**Figure 5.** Empirical eggNOG profile. (a) Pangenome spectrum. (b) Groups in the same COG category co-occur more than under shuffled category labels (phylogeny-controlled null).

detect that the structure exists, not to fit strong per-category coupling. A larger annotation is the first ask.

- **The path forward: COG categories as the coupling scaffold.** This resolves the structure problem cheaply — rather than inferring which families couple, take same-category groups as coupling blocks and fit the *strength* per category with `match_coupled` (exactly the setup this report validates). The library pieces are in place (`cooccurrence_summary`, `match_coupled`, the eggNOG parser); a coupled fit on a fully-annotated tree is the next milestone.



**Figure 6.** Internal co-occurrence cohesion per COG category (within-category minus global-between). Accessory/mobile functions are the cohesive modules; core housekeeping is not.